CSE 318 — Online 2

Section B1

Your local search stops as soon as it reaches a local optimum (no single flip improves the cut). But there might be a **better** local optimum nearby that you could reach if you "shook up" the solution a little and ran local search again.

Your task: After local search converges, apply a **perturbation** (randomly flip some vertices) and run local search again. Specifically:

1. Run your normal local search until it converges to a local optimum. Call this solution x_1 with cut value w_1.
2. **Perturb**: randomly select p vertices (without replacement) and flip each of them to the other side. This gives you a new (probably worse) solution.
3. Run local search again from this perturbed solution. Call the result x_2 with cut value w_2.
4. Return whichever of x_1 and x_2 has the larger cut value.

The parameter p controls how strong the perturbation is:

- p too small $\rightarrow$ you end up back at the same local optimum.
- p too large $\rightarrow$ you destroy the solution and the second local search starts from scratch.

Use $p = \lfloor 0.05 \times n \rfloor, \lfloor 0.1 \times n \rfloor, \lfloor 0.2 \times n \rfloor$ (flip 5%, 10% and 20% of the vertices).

What to run and report:

- Run GRASP with this "perturbed local search" on at least 6 benchmark graphs, for all three values of p.
- Run your original GRASP (local search without perturbation) on the same graphs.
- Use the same number of GRASP iterations for all.
- Report: best cut found and total time.